ZHAO BOXUAN

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PERSONAL PROFILE

Interdisciplinary design researcher exploring relationships between physical environments, bodily action, and interactive digital systems. Experience spans physical interactive installations, immersive exhibitions, real-time media, and district-scale AR research, connecting human-scale interaction with spatial organisation and system deployment. Developed an AR configuration decision framework, independently built a game using UE5 Blueprints, and led museum experience design through technical development, integrating interaction logic, physical structures, digital media, and implementation constraints. Research interests focus on how spatial conditions, action–feedback mappings, and multimodal cues shape exploration, understanding, and participation in embodied XR, with an emphasis on prototyping and empirical HCI research.

EDUCATION & COURSES

08/2022-06/2024 Master of Architecture

National University of Singapore

Coursework: Exhibition Curation and Spatial Design; AR Interaction Experience Design; Architectural Cinematics (UE5, Premiere Pro, After Effects); Digital Structures and Spatial Design (Rhino, Grasshopper, Lumion); Integrated Construction.

09/2016-06/2021 Bachelor of Architecture

Hunan University

Coursework: Computational Design; Parametric Structural Design; Building Physics, Building Materials, Environmental Psychology, Structural Mechanics. Design coursework ranked Top 5%.

THESIS RESEARCH

From Site Evidence to Situated AR Configurations: A Case-Derived Decision Framework for a Protected Historic District
Sole Author | Manuscript submitted to CAAD Futures 2027 (under review)

• **Decision Method |** Used Research through Design and retrospective cross-case comparison to translate heritage regulations, field evidence and visitor tasks into hard constraints, design preferences and unresolved variables. Compared AR with conventional and hybrid media, assessing whether spatial alignment added task-relevant value that justified its technical and operational demands.

• **Configuration Framework |** Developed a device-independent, two-axis classification of fixed versus movement-responsive viewing and shared versus personal visual streams, defining four situated AR configurations. Linked each to spatial registration, localisation, sightlines, interaction pacing, audiovisual synchronisation and equipment services, examining how viewing conditions and content distribution shape individual autonomy, collective participation and operational responsibilities.

• **Case Applications |** Applied the framework to four proposals in Xi' an' s Xiangzi Temple District: a shared courtyard theatre, personal-headset city-wall cinema, shared AR sightseeing bus and individual object-based exhibition. Mapped Han Xiangzi' s biography, alternative paths to immortality and later legends onto continuous, branching and dispersed spaces across three levels, connecting activities through visitor routes and equipment services.

• **Cross-Case Findings |** Compared how shared viewing concentrates support in display infrastructure and coordinated sessions, while personal viewing shifts demands towards calibration, independent audio and device management; movement-responsive activities require continuing localisation. Examined conventional wayfinding and headset-unavailability scenarios to identify retained functions, lost AR effects and revised service responsibilities, deriving a transferable media-decision procedure.

WORK EXPERIENCE

02/2025-02/2026 **Interaction Designer, VAVE Studio Shanghai** **Shanghai**

Exhibit Engineering | Detailed two installed interactive exhibits for Sinovac' s [My] Museum, coordinating fabrication and multimedia teams. For a suitcase-shaped display, specified aluminium framing, high-brightness LCD mounting, glare treatment, passive cooling and rear maintenance access within the exhibit structure.

Media Integration | Detailed a wristband-triggered human-history exhibit combining a physical terrain model, projected landscape imagery, and an animated river across six adjoining LCD panels. Modelled the terrain in Maya for fabrication, using rocks and fallen logs to conceal screen seams. Coordinated projection alignment, brightness and water-flow animations with the multimedia team to connect physical scenery with digital imagery.

Project Delivery | Independently developed successful concepts for Bailian' s ~1,000 m^2 CIIE booth and Design Shanghai' s ~150 m^2 Institute of Attention Economics. Completed ~70% of Bailian' s detailed design and served as lead designer in a two-person detailing team for Design Shanghai. Contributed to 10+ bids (five wins) and four built projects, directly coordinating construction on two.

Technical Adaptation | At Bailian, replaced a suspended PVC concept with steel framing, coordinating modular curved LED installation and a 45° mirror assembly. At Design Shanghai, redesigned embedded LCDs as suspended LEDs within modular aluminium frames, addressing mounting, pedestrian clearance, sprinkler coverage and textile movement through pre-build checks and on-site coordination.

SELECTED PROJECTS

TRUTH INCUBATOR (Interactive Media, Spatial Narrative & Game Development)

• **Conceptual Framework |** Compared information production and circulation across traditional and social media to formulate "information paralysis": a model in which competing narratives repeatedly capture attention and displace the original issue. Developed this model into the operational framework of a fictional "Content Farm."

• **Narrative Testing |** Classified misleading editing techniques through narrative coding of films and animation, alongside analysis of news footage. Applied these techniques to an original film and its manipulated re-edit, drawing on the Kuleshov effect. In a 47-participant test, 348 of 376 chapter-level answers (92.6%) matched the intended manipulated storyline, while the re-edit received a mean perceived-credibility rating of 4.62/5. Used these findings to inform the game' s editing rules.

• **Spatial Translation |** Translated content production, circulation and control into a layered spatial structure. Independently built environments and produced animations in Unreal Engine 5, using camera movement to reveal institutional operations and media power relations through spatial narrative.

• **Interaction Development |** Designed and independently implemented a news-editing game using UE5 Blueprints, including footage collection, an editing interface, scoring logic and publication feedback. Players construct three narratives from the same event and observe how editorial choices influence credibility and competition for attention within the game.

HEJIANG SOY SAUCE MUSEUM (Interactive Exhibition, Embodied Interaction & Technical Design)

• **Project Development |** Independently led content planning, spatial and interaction design, wayfinding, the successful bid and subsequent technical design development. Organised over ten interactive exhibits around visitor circulation, learning objectives, physical operations and media feedback, coordinating exhibit-level interactions with spatial and operational requirements.

• **Embodied Interaction |** Designed a Chishui River exploration corridor where walking, directing a handheld projector, scanning and dwelling reveal spatially anchored content across geology, climate and crops. Proposed 6DoF tracking, surface registration and projection correction to maintain imagery on simulated rock walls, with spatial audio, lighting and mist effects supporting narrative transitions and multisensory exploration.

• **Technical Integration |** Designed a polarisation-based exhibit revealing microbial animations through a handheld viewing filter. Compared LED, rear projection and LCD solutions, rejecting rear projection following lighting tests and cost assessment. Selected a modified LCD approach that transfers the front analysing polariser to the viewing filter, resolving the rectangular display–circular exhibit conflict through a masked aperture and concealed equipment housing. Specified polariser alignment, cooling and maintenance requirements for further testing.

TECHNIACL SKILLS

Interactive Development: Unreal Engine 5 (Blueprint visual scripting, interactive prototyping, real-time environments and animation).
Computational Design: Rhino, Grasshopper, Maya, SketchUp, 3ds Max; parametric modelling and spatial modelling.
Physical–Digital Integration: Interactive exhibit detailing, display and projection integration, sensor and hardware layout coordination, projection alignment, fabrication and maintenance planning.
Visual Media: Photoshop, Illustrator, After Effects, Premiere Pro; D5 Render, Enscape, V-Ray, KeyShot.

HONORS & ACADEMIC ACTIVITIES

Conference Paper | Contributed to Exploring the Restorative Potential of Public Green Space in University Residential Environments: A Case Study of HNU' s Tianma Student Apartments, selected for the 14th International Conference on Environment–Behavior Studies.
Selected Works | Marine Windpath recognised as outstanding student work at NUS and exhibited at ArchiVAL 2023; short film BOTTLE represented the NUS department at the Melaka International Film Festival.
Academic Honors | Hunan University Individual Scholarships (four consecutive academic years, 2016–2020); University-level Outstanding Student Leader (twice, 2017–2019).
Student Leadership | Head of Publicity, Hunan University School of Architecture Student Union; coordinated event communications and multimedia production, and managed the school' s WeChat account for two years, growing its audience to over 10,000 subscribers.